

Transportation Conditions and Inequality: Predicting Risk and Negative Youth Outcomes along the School-to-Prison Pipeline

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INTRODUCTION: WHY THIS MATTERS

According to the [American Civil Liberties Union](#), the school-to-prison pipeline refers to a historic and existing system within the United States in which children and teenagers within the American education system are funneled out of the nation's public schools and into juvenile and criminal systems, disproportionately impacting youth of historically-marginalized and underserved communities.

Research has explored relationships across a wide range of contributing factors that intensify youth's likelihood to fall into the school-to-prison pipeline, with paying particular attention to the role of systemic, economic, environmental, and social attributes, such as harsh disciplinary policies, police presence on school campuses, lack of trauma-informed and/or culturally-competent resources school support, and heightened suspension and drop-out rates. According to the [National Bureau of Economic Research](#), students assigned to schools with suspension rates one standard deviation higher than average were 15%-20% more likely to be arrested or incarcerated as adults, more likely to drop out, and less likely to enroll in a 4-year college.

Not only do institutional responses to student behavior shape whether students stay engaged with school or are pushed outward, but students' lived conditions along the lines of race, class, income, and other demographic indicators have become driving factors that subject them to greater susceptibility of the school-to-prison pipeline, even before stepping foot onto the school. The [ACLU](#) states that Black and Brown students and students with disabilities are particularly vulnerable. Pipeline vulnerability is additionally linked to students living and/or attending schools in lower-income areas, intertwined with school quality, neighborhood disadvantage, race, and class-based stigma and bias. Across numerous studies, the school-to-prison pipeline is underscored by social scientists as a systemic issue, rooted in and manufactured overwhelmingly by the conditions of a youth's environment (physical, educational, socioeconomic, etc.) rather than solely individual student behavior, while especially impacting at-risk children and transition-aged youth of society's most vulnerable demographics.

This research project further analyzes the contributing factors to the school-to-prison pipeline by paying particular attention to the role of **transportation conditions and inequality** in how it might shape or predict school-based risk indicators (e.g. chronic absenteeism, poor academic performance, suspension/drop-out, school behavioral and exclusionary discipline) that frequently result in youth entering juvenile or criminal systems. More specifically, the project hones in on the outcome of **poor academic performance**, seeking to clarify the relationship between mobility access and transportation safety and how well students perform in school, which is additionally attributed to trends of chronic absenteeism and potential risk of suspension or drop-out. As a **computational spatial analysis** of unequal transportation and mobility environments in **San Jose and Stockton, California**, this project utilizes data of these two cities' public high schools, with transportation and mobility as a potential predictive lens, to help answer **where and how** (geographically, environmentally, and socioeconomically) a certain risk indicator (i.e. poor academic performance) on the school-to-prison pipeline happens. This project will utilize computational social science and data methods executed in R,

with code cited from this public [Github repository](#) created for this project. All data sets used in my research are also available in that repository.

With ongoing structural barriers that not only prevent youth educational success and positive life outcomes, but actively funnels the United States' young people into cycles of harm, violence, and marginalization, understanding the key risk indicators of the school-to-prison pipeline, and the condition that manufacture such, is imperative for developing comprehensive solutions that challenge the pipeline early-on, as well as interrupt its perpetuation of systemic disparities that prevent all families, youth, and communities from thriving.

Goals of this Research

This research intends to...

1. Predict

Which school-neighborhood zones have a higher risk of poor academic performance and potentially the school-to-prison pipeline based on transportation inequality and unsafe transportation-mobility environments.

2. Test

The relationship between transportation, mobility, and risk indicators (i.e. poor academic performance in schools) and its intersection with income.

3. Analyze

Transportation and mobility conditions and inequality to build preventative measures against negative youth outcomes on the school to prison pipeline.

Core Question

How can we use mobility inequalities and unsafe transportation environments to predict academic performance as a pipeline risk indicator in Stockton and San Jose public high schools?

SCOPE OF DATA

Why These Cities?

This research project chose to focus specifically on San Jose and Stockton, California as meaningful case studies because these cities' data and further information generally reflected the following key features:

- **Students largely go to school where they live:** In contrast to high schools in San Francisco and New York City where students apply to their high schools across the broader city region, resulting in enrollment that often differs from the residential make-up surrounding the school, the vast majority of San Jose and Stockton public high schools, including magnet and charter public schools, follow the traditional neighborhood-based system where school enrollment is assigned based on place of residence. As a result, transportation conditions surrounding school campuses can more accurately relate to conditions and outcomes for the students actually attending the school, whether absenteeism or school performance.
- **There are strong publicly available data sets,** likely due to the fact that public schools in these cities have relatively higher enrollment numbers and are associated with large school districts that have trackable, accessible information available, which is additionally related to higher-level entities, such as the CA Department of Education, where data collection is robust and thorough across the state.
- **Visible inequalities and disparities:** Across both San Jose and Stockton exists visible infrastructure inequality that shapes the local environmental and structural landscape of the population. In addition, these cities have large lower-income and youth-of-color populations, which reflects the disproportionately-impacted demographics along the school-to-prison pipeline as reflected in prior research. The research also intends to analyze how transportation and mobility inequalities and poor academic performance map across the existing, measurable disparities in these cities, such as intense wealth inequality in San Jose and a history of economic divestment that has particularly impacted Stockton's immigrant/refugee populations.

Standardized Unit of Analysis

For this project, I will be analyzing using **school-neighborhood zones**. Each of these zones, or buffers, will consist of 1 high school and a surrounding 1-mile radius around the address of the high school. The 1-mile radius accounts for the surrounding neighborhood transportation and mobility environment, encompassing car and transit routes and other factors within the environmentally-structurally influential zip codes, in the vicinity of the school analyzed. There are a total of 46 school-neighborhood zones analyzed in this research. A flowchart representation of this unit of analysis is depicted below:

School

↓ embedded within ↓

Neighborhood transportation / mobility environment within a 1.0 mile radius

↓ associated with ↓

Poor academic performance vulnerability

This research project draws on data from a **total of 46 high schools** (32 in San Jose, 14 in Stockton). Data is compiled and analyzed both separately across each city and combined.

These 46 high schools were selected from the [California Department of Education's statewide school directory](#), which I manually filtered to search for public high schools (serving up **grade levels including 9th and up to 12th**), including public magnet and charter schools, that were located within the cities of San Jose and Stockton. Information including school address, district, zip code, city, charter-magnet-traditional model, grades served, county, and school district could be exported from the directory into an Excel sheet used for analysis. Two separate Excel sheets for [San Jose](#) and [Stockton](#) were created.

In order to conduct analysis and create these buffers as standardized units for this research project, I first geocoded (converting from addresses into geographic coordinates) each of the 46 school addresses, primarily using Census geocoder and OpenStreetMap when needed. These coordinates were then compiled and saved in a [csv file](#). *Code for this process is viewable on lines 9 through 60 in the .Rmd file in [Github](#).* I then created 1-mile buffer zones around each address coordinates using the sf package and the map points. *Code for this process is viewable on lines 62 through 84 in the .Rmd file in [Github](#).* Lastly, to get information useful for later analysis, I estimated the population within each of these 1 mile buffer zones by gathering data from the [2019-2023 ACS](#) census tract populations for Santa Clara and San Joaquin Counties, via an API key. Through areal interpolation, I intersected the buffers with census tracts to find portions of overlap and which portions match between the two units, therefore estimating [population](#). *Code for this process is viewable on lines 86 through 146 in the .Rmd file in [Github](#).*

VARIABLES AND FORMULA

Independent Variables

The independent variable data for this research project is organized into and analyzed as two distinct categories: mobility access and transportation safety. As a result, I created two indexes, each regarded as its own broader independent variable that encompasses other smaller, more-specific data sets.

The Mobility Access Index (MAI) looks at the access to different transportation methods within a school's buffer zone, exploring the question, "Can students physically get where they need to go? (i.e. school)". Data sets analyzed under this index include the percentage of households in the buffer zone without access to a vehicle, the distance to the nearest transit stop, and the availability of school-based transportation methods (e.g. school bus systems).

The Transportation Safety Index (TSI) looks at the safety of the area within the school's buffer zone that is relevant to the transportation routes a student needs to take when going to

school, pressing “Is it even safe for a student to get there?”. Data sets analyzed under this index include the rate of exposure to violent crime and other relevant crime events in the area and the density of crashes involving bikes and pedestrians.

Each of these comprehensive indices are analyzed as an independent variable altogether in MAI or TSI, associated with the dependent variable.

Dependent Variables

As I focus on **poor academic performance** as a risk indicator on the school-to-prison pipeline and how transportation and mobility conditions impact performance in school, this research project will use **standardized test scores** as measurable outcome data across California high schools. More specifically, this project draws on data from the **CAASPP ELA and Math Exams**, the standardized benchmark exam taken by all California high school students in the 11th grade. To measure poor academic performance, I will be looking at the **% of students who have not met the passing standard**, or % unsatisfactory.

I chose this dependent variable as there is guaranteed available data at the school-site level from the statewide CAASPP data set, linking well with the individual school-specific buffer zones. Although mobility access and transportation safety more directly correlate with chronic absenteeism, the majority of schools had no chronic absenteeism available and school-site available data was far more accessible for test scores. I regarded test scores as the next most accurate data set, the logic being that absence in school strongly relates to negative impact on school learning and academic results, as proven by various existing studies.

Poor academic performance, as reflected in test scores, links to transportation barriers on one end and long-term school exclusion on the other, in addition to being measurable and trackable statistically.

Calculating Risk

As I aim to explore how disparities in mobility access and transportation safety across areas might impact academic performance, I **established a formula** using the Mobility Access Index and Transportation Safety Index as a **calculation for “risk,” or the risk of a school-to-prison pipeline indicator, such as poor academic performance**. Risk will be seen as a direct dependent variable from the MAI and TSI, and used later in this project to compare with actual CAASPP test scores.

Risk is quantified as a **“Risk Score,”** along with scores based on the Mobility Access Index and Transportation Safety Index.

The **MAI** will be measured on a score scale of **0.0 to 1.0**, with 0.0 indicating very poor mobility access and more mobility access inequality, and 1.0 indicating excellent mobility access with lesser mobility access inequality.

The **TSI** will be measured on a score scale of **0.0 to 1.0**, with 0.0 indicating a very unsafe transportation environment and more transportation safety inequality, and 1.0 indicating a very safe transportation environment with lesser transportation safety inequality.

On the same scale as MAI and TSI, Risk Score will be measured from 0.0 to 1.0, with 0.0 indicating the lowest risk and 1.0 indicating the highest risk. In order to calculate Risk Score, the average of MAI and TSI is subtracted from 1. In other words, we will use the formula of:

$$\text{Risk} = 1 - (\text{MA} + \text{TS}) / 2$$

This formula will be used in later stages of data processing.

Examples of calculations using MAI, TSI, and Risk Score are provided below:

Excellent School-Neighborhood Transportation/Mobility Environment

Variable	Value
MAI	0.9
TSI	0.8

Calculate Risk:

$$1 - (0.9 + 0.8) / 2$$

$$= 1 - 0.85$$

$$\text{Risk} = 0.15$$

Low risk.

Poor School-Neighborhood Transportation/Mobility Environment

Variable	Value
MAI	0.2
TSI	0.1

Calculate Risk:

$$1 - (0.2 + 0.1) / 2$$

$$= 1 - 0.15$$

$$\text{Risk} = 0.85$$

High risk.

Again, as shown, we use Risk Score as a direct dependent variable of the Mobility Access Index and Transportation Safety Index through an input-output equation. The main question we want to explore in this research, now, is can we use **Risk Score** (based off of MAI and TSI) as an **accurate predictor of poor academic performance**? Later phases in this research will compare risk scores directly with poor academic performance to analyze relationships.

DATA PROCESSING

Data used in this research comes from the most recently available set of data across all sources.

Note that the list of all 46 public high schools analyzed and their information has already been gathered from the CA Department of Education statewide school directory, as described in the above [Scope of Data](#) section.

CAASPP ELA and Math Score (% Unsatisfactory)

I accessed the school-level data for the 2024-2025 school year from the [CAASPP database](#), manually filtering my search for only the ELA and Math score data for all public schools (all grade levels) in Santa Clara and San Joaquin County. In R, I additionally filtered to gather data only for high schools and only for those located in the cities of [San Jose](#) and [Stockton](#), matching the 46 found on the CDE directory and only looking at the percentage of students who scored an unsatisfactory score for the CAASPP ELA and Math exam for each school. This information was then compiled into two separate data sets, one for each city, displayed in both % format and 0.0-1.0. *Code for this process is viewable on lines 148 through 220 in the .Rmd file in [Github](#).*

Mobility Access Index (MAI)

The Mobility Access Index looks at data sets across both public transportation infrastructure and institutional transportation support provided to students through their school districts, understanding the various factors that contribute to mobility access within an area. The MAI is the **average of the below data** on the 0.0 to 1.0 scale.

School-based Transportation Availability:

This looks at the availability of school buses and school bus routes for each school-neighborhood buffer available via the school district serving that school. This information was available and manually gathered from the websites of the school districts represented in the 46 high schools across San Jose and Stockton. This included [Campbell Union High School District](#), [East Side Union High School District](#), [Fremont Union High School District](#), [San Jose Unified School District](#), [Santa Clara County Office of Education](#), [Lincoln Unified School District](#), [Lodi Unified School District](#), [Manteca Unified School District](#), and [Stockton Unified School District](#).

To measure school-based transportation availability on a 0.0 to 1.0 positive scale, I created the following rubric criteria levels based on patterns that were prevalent across the school districts:

- **0.0 - no access**
 - No bus service
- **0.25 - low access**
 - Transportation only for certain demographics (e.g. special education students, income)
 - AND must still apply
 - esp. also must meet geographic boundaries
- **0.5 - middle access**
 - Service if meet geographic boundaries AND special education - OR - if meet geographic boundaries AND certain income thresholds
 - Esp if students must still apply for bus access
- **0.75 - more access**
 - Service if meet geographic boundaries
 - Application
- **1.0 - broad access**
 - Most students can ride (i.e. no application)
 - District operates regular transportation routes
 - Transportation is a standard service

These scores were compiled with the rest of the school data for all 46 high schools, separated into a [San Jose](#) and [Stockton](#) data set.

Vehicle Access in a Household:

For each of the 46 school-neighborhood 1-mile buffer zones, I gathered the percentage of households within that buffer zone that did not have access to a vehicle. This was done in R using the [2019-2023 ACS census data](#), which was gathered through an API key.

To measure this on the correct positive scale from 0.0 to 1.0, I created a **vehicle access sub-score** calculated from **1.0 - (Raw percentage of households without vehicles) / 100**, with a vehicle access score of **0.0** indicating that no households within that buffer zone has vehicle access and **1.0** indicating vehicle access for every household in that buffer zone.

Code for this process is viewable on lines 292 through 340 in the .Rmd file in [Github](#).

Distance to the Nearest Transit Stop:

To find the distance between the school address and the nearest transit stop for all 46 high schools, I first downloaded city-based transit stop information through the [General Transit Feed Specification](#) open data source, available for [San Jose](#) and [Stockton](#) as txt files. The data specifically represents bus stops, light rail platforms, and light rail stations from Santa Clara Valley Transportation Authority (in San Jose) and bus stops from San Joaquin Regional Transit (in Stockton). In R the sf package map points, I then read each school's coordinates to get distance (in meters) to the nearest stop, converting it into miles and normalizing the distance

on a 0.0 to 1.0 scale. *Code for this process is viewable on lines 222 through 279 in the .Rmd file in [Github](#).*

To measure this on the correct positive scale from 0.0 to 1.0, I created a **transit distance sub-score** calculated from **1.0 - (Distance to Nearest Transit Stop, normalized)**, with a transit distance score of **0.0** indicating a very far distance to the nearest stop and **1.0** indicating a very close distance to the nearest stop, all within the buffer zone. Essentially, the greater the value of the transit distance score, the less distance from school to the nearest transit stop. This information was all calculated and compiled in respective spreadsheets for [San Jose](#) and [Stockton](#).

Transportation Safety Index (TSI)

The Transportation Safety Index looks at data sets across crime rates and crash density, understanding the various factors that contribute to transportation safety within an area. The TSI is the **average of the below data** on the 0.0 to 1.0 scale.

Reported Crime Exposure Rate:

To get the reported crime exposure rate within each school-neighborhood buffer zone, I manually searched for each of the 46 school addresses into [crimemapping.com](#), the top recommended source from the cities' police departments, extracted on a frequent basis from police records. I applied the platform filters to allow me to gather information on the number of crime events within the 1 mile radius of the school-neighborhood buffer zone recorded in the past 4 weeks, manually compiling the data into spreadsheets for San Jose and Stockton schools. Types of crime considered included assault, arson, robbery, homicide, burglary, theft/shoplifting, vandalism, missing persons, and kidnapping, considering events that would be the most influential to and reflective of safety for students going to school.

I then ran calculations on the spreadsheet to get the **crime rate per 1,000 residents** within the 1 mile buffer zone, using [population data](#) found previously and **(number of total crimes / population) × 1,000**. Measuring this on the **normalized scale** from 0.0 to 1.0, I also created a **crime safety sub-score** calculated using **1.0 - (crime rate, normalized decimal)**, with a crime safety score of **0.0** indicating the least safety with the highest crime rate and **1.0** indicating the most safety with the lowest crime rate, all within the buffer zone. This information was calculated and compiled in respective spreadsheets for [San Jose](#) and [Stockton](#).

Crash Density (involving Bike and Pedestrian):

To get the crash density within each school-neighborhood buffer zone, I manually searched for each of the 46 school addresses into the [Berkeley TIMS](#) Safe Routes to School Crash Data Map, which would report and display the number of crash records out of 100 within a ½ mile radius of the school, from 2018 to 2024. This information was manually gathered.

To measure this on the correct positive scale from 0.0 to 1.0, I created a **safety from crash sub-score** calculated from **1.0 - (Raw number of crash events) / 100**, with a safety from crash score of **0.0** indicating very unsafe and **1.0** very safe in that buffer zone. This information was

calculated and compiled in respective spreadsheets for [San Jose](#) and [Stockton](#).

Joining Data and Calculations

After gathering data across various resources for the outcome variable, Mobility Access Index, and Transportation Safety Index, I joined the data sets together in R to compose two comprehensive data sets, one for San Jose, another for Stockton, displaying all relevant data for the 1-mile school-neighborhood buffer zone.

With the MAI, TSI, and Risk Score formulas [explained above](#), I calculated and appended these three additional data pieces for each high school into the large comprehensive data set in R. *Code for this process is viewable on lines 281 through 437 in the .Rmd file in [Github](#).*

```
San Jose schools: 32
Stockton schools: 14
Columns: School | District | Street Address | Public Yes/No | Magnet Yes/No | Charter Yes/No |
Low Grade | High Grade | City | transport_rating | transit_score | vehicle_access_score |
crime_safety_score | crash_safety_score | caaspp_unsatisfactory | mobility_access_index |
transportation_safety_index | risk_score
```

Demographic Data

To get information useful for later analysis, I will gather **demographic information** within each of the 1-mile school-neighborhood buffer zones using the [2019-2023 ACS](#) census data, such as that across zip codes, for **Santa Clara** and **San Joaquin Counties**, via an **API key**. Additional demographic information was considered and joined onto sj_data and st_data throughout the course of this research project.

MODELS, VISUALIZATIONS, AND INSIGHTS

To interpret our cleaned and processed data, I employed choropleth maps, scatterplots, tables, and simple linear regression modeling in order to understand the relationships between mobility access, transportation safety, risk, and poor academic performance.

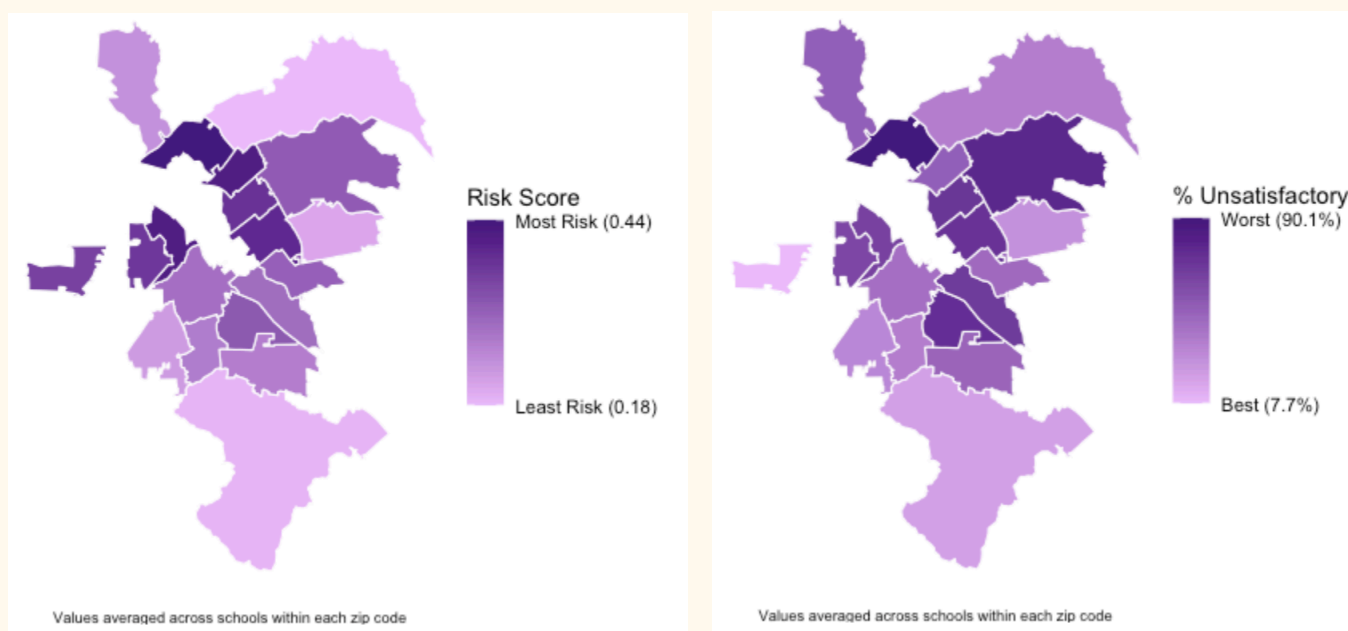
A **linear regression model** aims to show the relationship between an outcome and one or more explanatory variables by fitting a straight line through the data of the form $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$. One key output of linear regression is the **slope** (β_1), which quantifies how much the outcome variable is expected to change for each one-unit increase in the predictor (independent variable, or x). The **intercept** (β_0) is the line's baseline value. y_i is the **predicted value of the outcome**, or dependent, variable. ϵ_i is the error term, or residuals.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

A linear model can be used to gain insight into **how strongly** and in **which direction** an independent and dependent variable are related, as in, how much of a **change in one variable** results in a certain amount of **positive or negative change in the other**. In addition, a linear regression model can be used for prediction of the **value of a dependent variable** given the **value of the independent variable**. As a data model, the linear regression model allows each variable to get an interpretable coefficient, and is well-suited for a smaller data set of 46 schools

CAASPP ~ Risk Score Predictions

Below are two choropleth graphs of **San Jose**, displaying the ranges of **Risk Scores** on the left and the ranges of **% Unsatisfactory CAASPP Scores** on the right. **Darker colors** signify **higher values**, for both percentages and decimals. The values are averaged across schools within each zip code for better visual display. We see that the mapping of Risk Scores exhibits similar patterns to % Unsatisfactory CAASPP, with darker colors (higher values) clustered in the **central** and slightly **northeast** parts of the city. *Code for these maps is viewable on lines 532 through 623 in the .Rmd file in [GitHub](#).*



To more accurately understand the relationship between Risk Score and % Unsatisfactory CAASPP in San Jose, a simple linear regression model can be used to see how Risk Scores might predict % Unsatisfactory CAASPP Scores in the form **caaspp_unsatisfactory** = β_0 + $\beta_1(\text{Risk Score})$. *Code for this model is viewable on lines 1473 through 1513 in the .Rmd file in [GitHub](#).*

```
--- SAN JOSE: CAASPP ~ Risk Score (n = 31 ) ---
parsnip model object
```

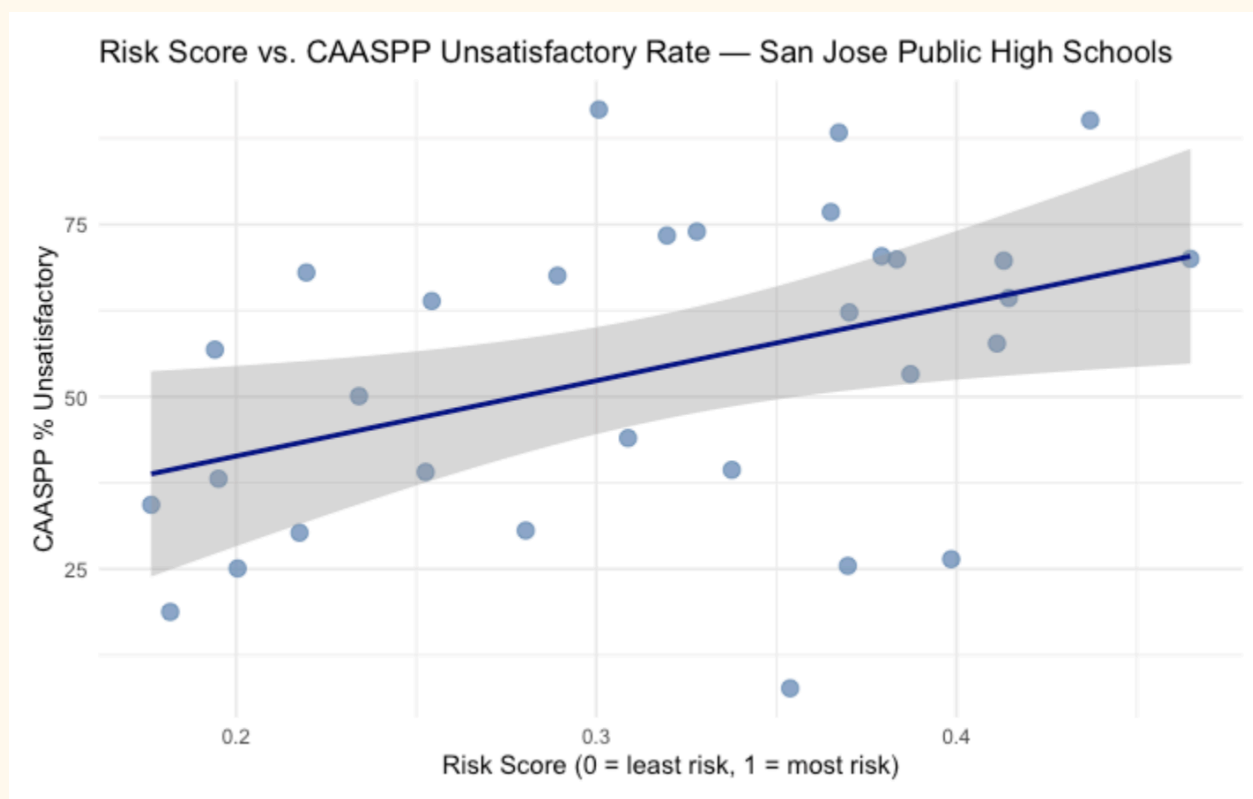
```
Call:
stats::lm(formula = caaspp_unsatisfactory ~ risk_score, data = data)
```

```
Coefficients:
(Intercept)  risk_score
    0.1947      1.0951
```

The **intercept of 0.195** indicates that a hypothetical San Jose high school with a **Risk Score of 0** is predicted to have **~19.5% of students** fail to meet the CAASPP standard, the baseline anchor for this form. The positive slope of **+1.095** indicates that every positive **+0.1** in Risk Score predicts **+0.11 (or plus 11%)** more students that fail to meet standard on the CAASPP. San Jose supports the hypothesis that a higher Risk Score **relates positively** to worse academic outcomes, as the **slope remains positive and steep**.

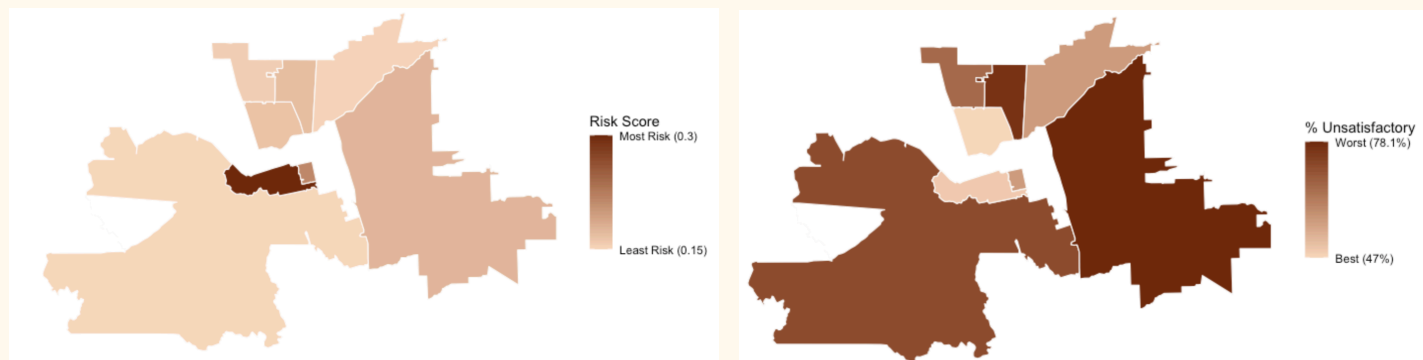
Using the data points for high schools in San Jose, I created a scatter point to represent the data with a line of best fit. The graph displays a positive slope, with the increase in Risk Score positively related to an increase in % Unsatisfactory CAASPP. *Code for this plot is viewable on lines 1279 through 1299 in the .Rmd file in [GitHub](#).*

With $R^2 = 0.17$, this means that **~17%** of the variation in the outcome variable can be explained by the model. R^2 values closer to 1 indicate a better fit model.



Moreover, below are two choropleth graphs of **Stockton**, displaying the ranges of **Risk Scores** on the left and the ranges of **% Unsatisfactory CAASPP Scores** on the right. Like San Jose, **darker colors** signify **higher values**, for both percentages and decimals, and the values are averaged across schools within each zip code for better visual display. While the mapping of Risk Scores exhibits less similar patterns to % Unsatisfactory CAASPP in Stockton compared to

San Jose, the darker colors (higher values) still seem to be clustered in the **east** side of the city. The variation for Stockton can also be due to the fact that the city has a significantly smaller sample size of 13 high schools (1 suppressed out of the 14 due to unavailable CAASPP data), compared to San Jose's 32. *Code for these maps is viewable on lines 778 through 869 in the .Rmd file in [GitHub](#).*



To more accurately understand the relationship between Risk Score and % Unsatisfactory CAASPP in Stockton, we will again use a simple linear regression model in the form **caaspp_unsatisfactory** = $\beta_0 + \beta_1(\text{Risk Score})$. *Code for this model is viewable on lines 1473 through 1513 in the .Rmd file in [GitHub](#).*

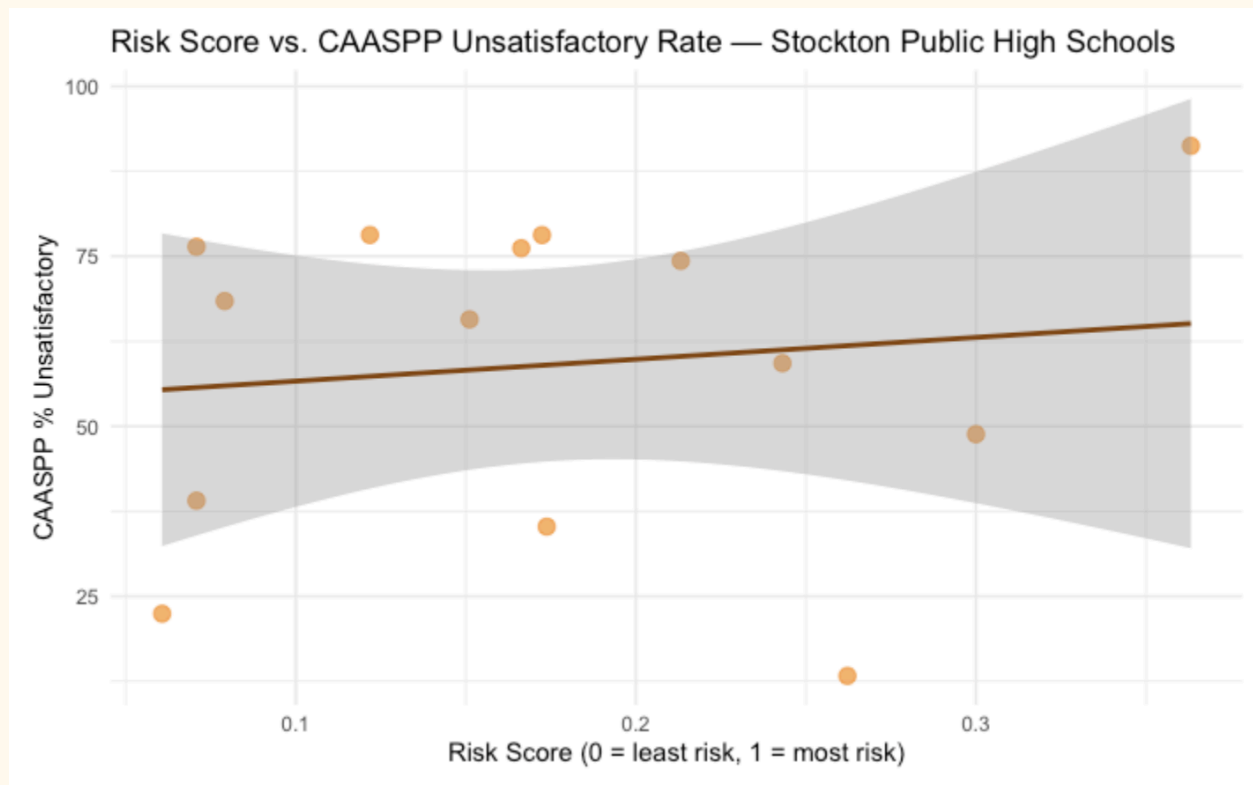
```
--- STOCKTON: CAASPP ~ Risk Score (n = 14 ) ---
parsnip model object

Call:
stats::lm(formula = caaspp_unsatisfactory ~ risk_score, data = data)

Coefficients:
(Intercept)  risk_score
    0.5343      0.3216
```

The **intercept of 0.534** indicates that a hypothetical Stockton high school with a **Risk Score of 0** is predicted to have **~53.4% of students** fail to meet the CAASPP standard, the baseline anchor for this form. The positive slope of **+0.322** indicates that every positive **+0.1** in Risk Score predicts **+0.03 (or plus 3%)** more students that fail to meet standard on the CAASPP.

A scatter point to represent the data for Stockton is below. While the graph displays a positive slope, the slope is much flatter for Stockton compared to San Jose, representing a weaker relationship. *Code for this plot is viewable on lines 1364 through 1384 in the .Rmd file in [GitHub](#).*



With $R^2 = 0.02$, this means that only ~2% of the variation in the outcome variable can be explained by the model. As a result, the relationship between Risk Score and % Unsatisfactory CAASPP Score, with the given data of only 13 schools, is **weak and noisy for Stockton**.

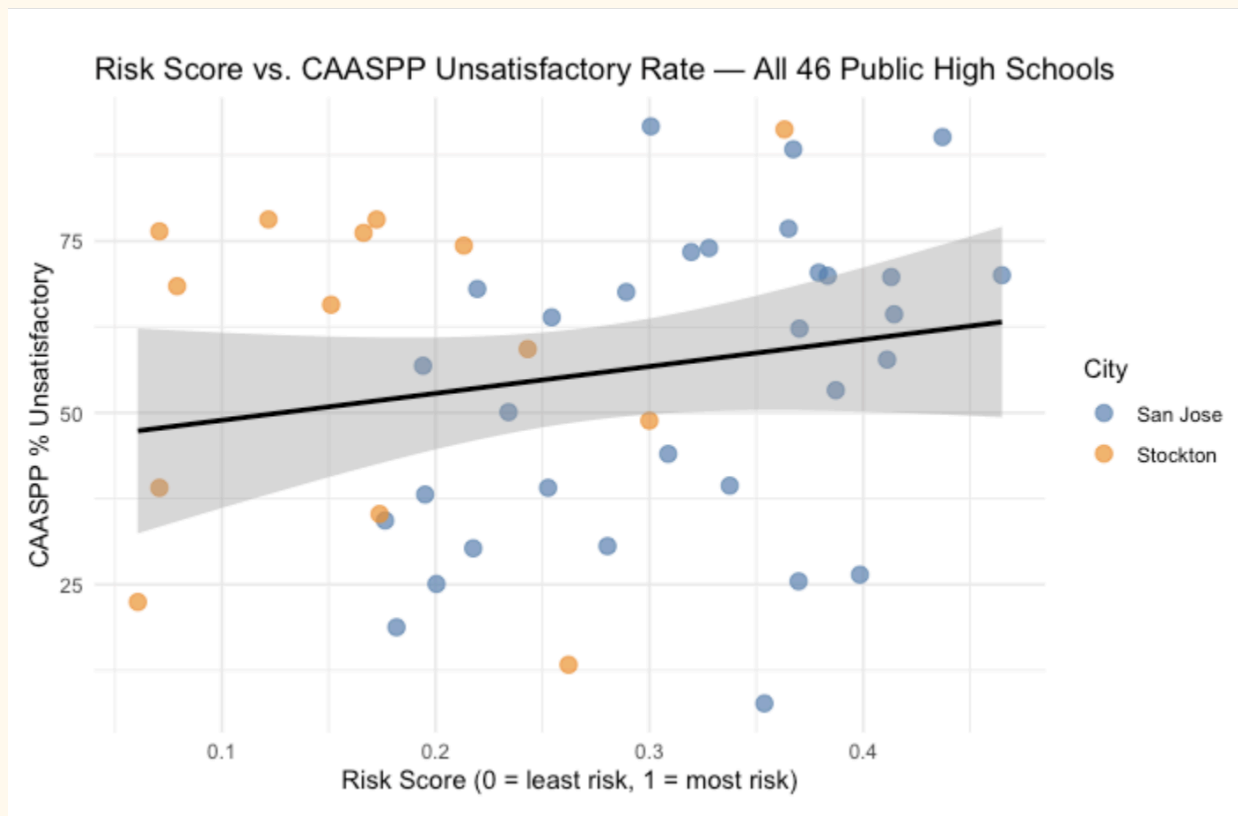
Across both San Jose and Stockton, a hypothetical high school with a **Risk Score of 0** is predicted to have **~44.9% of students** fail to meet the CAASPP standard. The positive slope of **+0.391** indicates that every positive **+0.1** in Risk Score predicts **+0.04 (or plus 4%)** more students that fail to meet standard on the CAASPP. However, this is probably not accurate to predict outcomes comparatively across actual high schools in these two cities, due to great variation in such a small data set.

```
--- BOTH CITIES: CAASPP ~ Risk Score (n = 45 ) ---
parsnip model object
```

```
Call:
stats::lm(formula = caaspp_unsatisfactory ~ risk_score, data = data)
```

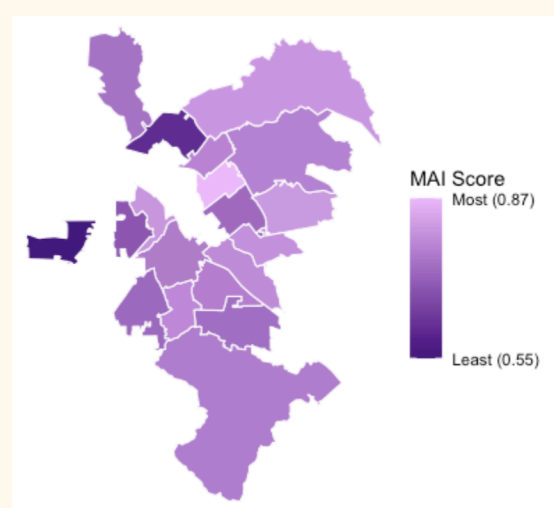
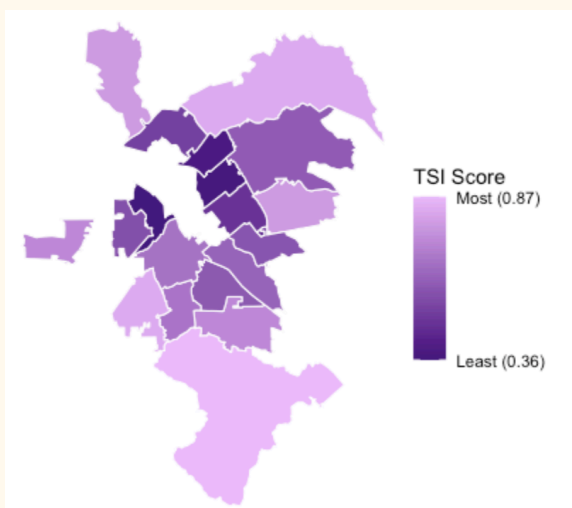
```
Coefficients:
(Intercept)  risk_score
    0.4498      0.3919
```

Below is a scatter plot with data points from both cities. With $R^2 = 0.04$, this means that only ~4% of the variation in the outcome variable can be explained by the model



CAASPP ~ MAI and TSI Predictions

Below are two choropleth graphs of **San Jose**, displaying Transportation Safety Index and Mobility Access Index (left to right). Darker colors indicate less safe environments and lesser mobility access, respectively. Mobility Access Index seems to be the variable that is the least similar to all other choropleth maps displayed for San Jose so far. Code is viewable on lines 439 through 530 in the .Rmd file in [GitHub](#).



A linear regression model is displayed below for further interpretative insight, displaying Mobility Access Index and Transportation Safety Index in San Jose. *Code for this model is viewable on lines 1515 through 1561 in the .Rmd file in [GitHub](#).*

```
--- SAN JOSE: CAASPP ~ MAI + TSI (n = 31 ) ---
parsnip model object

Call:
lm(formula = caaspp_unsatisfactory ~ mobility_access_index +
    transportation_safety_index, data = data)

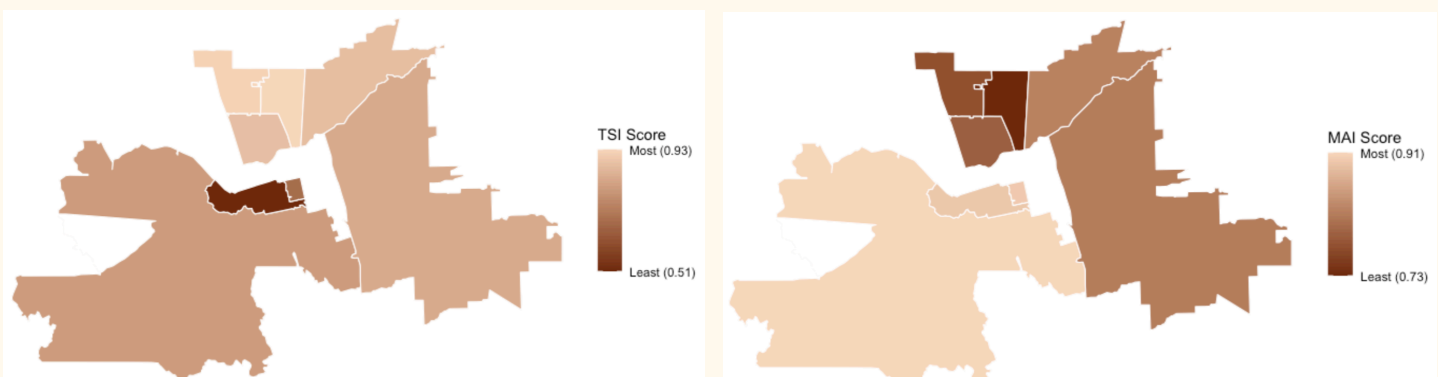
Coefficients:
            (Intercept)      mobility_access_index transportation_safety_index
                0.7098                  0.3662                  -0.7305
```

Holding Mobility Access Index constant, a **+0.1** change in the **Transportation Safety Index** predicts a **-0.073 or -7%** in the percent of students who fail to meet CAASPP score standards. Safer transportation environments are positively related to fewer students failing. On the other hand, holding Transportation Safety Index constant, a **+0.1** change in the **Mobility Access Index** predicts a **+0.036 or +3.6%** in the percent of students who fail to meet CAASPP score standards. This goes against the prediction, as it displays that higher mobility access predicts **worse** academic performance.

The coefficient for Transportation Safety Index is roughly **twice the size** of that for Mobility Access Index, indicating that **transportation safety is the dominant variable** predicting % Unsatisfactory CAASPP Score in San Jose.

As for the flipped Mobility Access Index, a potential explanation could be that schools in areas with the highest mobility access tend to be in denser urban areas that carry other disadvantages, such as income disparity. When controlled for transportation safety, mobility access “worsens scores.” When in reality, this display can be very likely due to the two predictors actually overlapping in real life.

For Stockton, two choropleth graphs displaying Transportation Safety Index and Mobility Access Index (left to right) are shown below. There appears to be no consistent pattern in the mapping of these two variables. *Code for these maps is viewable on lines 685 through 776 in the .Rmd file in [GitHub](#).*



A linear regression model is displayed below for further interpretative insight, displaying Mobility Access Index and Transportation Safety Index in San Jose. *Code for this model is viewable on lines 1515 through 1561 in the .Rmd file in [GitHub](#).*

```
--- STOCKTON: CAASPP ~ MAI + TSI (n = 14 ) ---
parsnip model object
```

Call:

```
stats::lm(formula = caaspp_unsatisfactory ~ mobility_access_index +
  transportation_safety_index, data = data)
```

Coefficients:

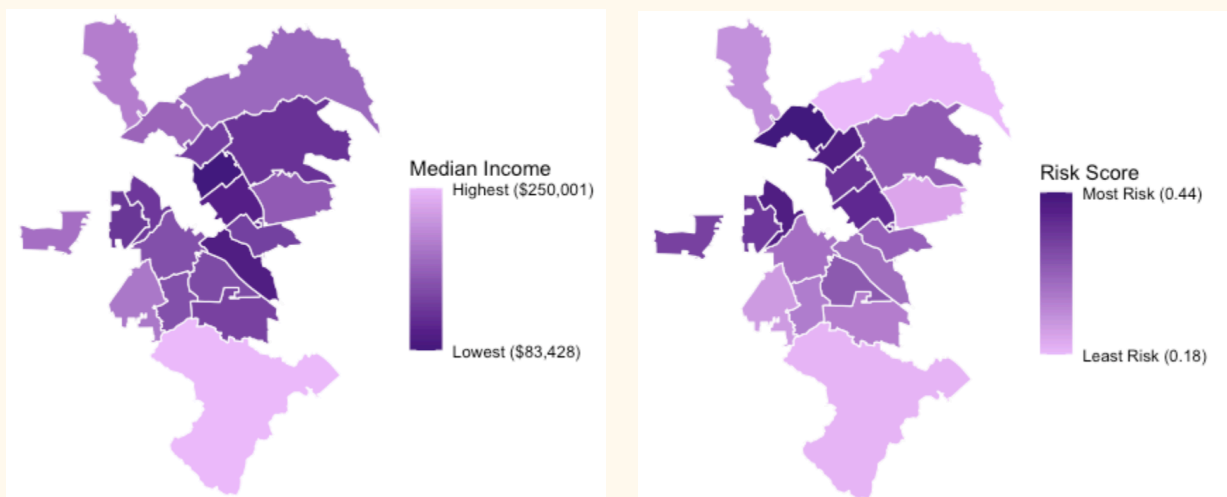
(Intercept)	mobility_access_index	transportation_safety_index
0.5568	0.2434	-0.2057

Holding Mobility Access Index constant, a **+0.1** change in the **Transportation Safety Index** predicts a **-0.021 or -2%** in the percent of students who fail to meet CAASPP score standards. As with San Jose, safer transportation environments are positively related to fewer students failing in Stockton. Holding Transportation Safety Index constant, a **+0.1** change in the **Mobility Access Index** predicts a **+0.024 or +2.4%** in the percent of students who fail to meet CAASPP score standards. The same explanation for this flip in San Jose can likely be applied to Stockton. Mobility Access Index seems to be the slightly more significant variable in Stockton compared to San Jose, but this is less meaningful due to the smaller data set.

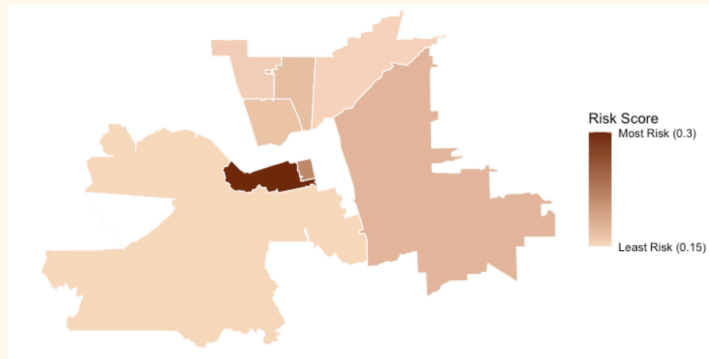
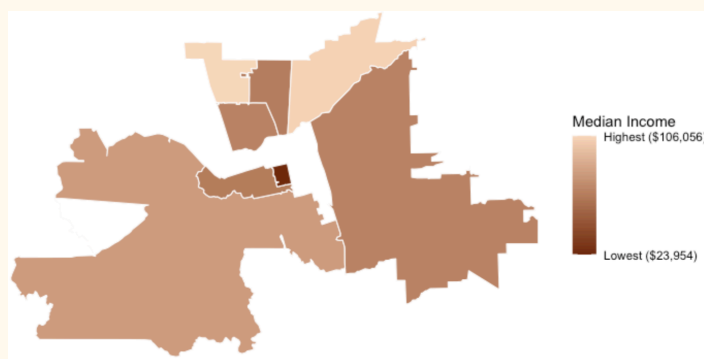
Risk ~ Income Predictions

I additionally wanted to focus on how systemic factors such as income could be used as another layer of prediction for risk, and therefore potentially poor academic performance. Below are choropleth maps comparing Income vs. Risk in San Jose and Stockton. **Higher-risk** and **lower-income** areas seem to cluster in the **central** and **slightly northeast** parts of the city for **San Jose**, while **Stockton** shares some overlap in the **east** and **a little lesser in the center**.

San Jose:



Stockton:



Code for these above maps are viewable on lines 625 through 683, 532 through 623, 871 through 929, and 778 through 869 in the .Rmd file in [GitHub](#).

To view this relationship more precisely, I created tables in R that listed the **Top 5 Most At-Risk** and **Top 5 Least At-Risk High Schools** for both **San Jose** and **Stockton**. In addition to displaying school name, ranking, and Risk Score, each row also displayed that **school-neighborhood buffer zone's median household income** in comparison to the **median household income for the rest of the city**. Income information was accessed through the [2019-2023 ACS](#) census data via an API key, and income for each buffer was found via census tracts. Through this method, I intended to understand how income inequality, whether or not a school-neighborhood is higher or lower in income relative to the city average, relates to greater or lesser risk for that school-neighborhood, likely implying better transportation safety, better resources, and other systemic advantages linked to income. *Code for this process is viewable on lines 931 through 997 and 1110 through 1171 in the .Rmd file in [GitHub](#).*

In **San Jose**, where the average median household income is **\$141,565**, **4 out of 5** of the **Top 5 most at-risk schools** are located in school-neighborhoods with lesser household income compared to the city average, ranging from **-\$1,687** to **-\$52,087**. For the one higher-risk school-neighborhood that lists a median income of **+\$20,427**, the school, **Opportunity Youth Academy**, is actually a **specialized academic program** for students who have dropped out of school previously and/or are disproportionately impacted by poverty, but is physically located in a higher income address.

In addition, **5 out of 5** of the **Top 5 least at risk-schools** are located in school-neighborhoods with higher household income compared to the city average, ranging from **+\$31,765** to **+\$98,463**.

--- San Jose: Top 5 Most At-Risk ---

A tibble: 5 × 4

School <chr>	`Risk Score` <dbl>	`Buffer Income` <chr>	`vs. City Median` <chr>
1 ACE Charter High	0.465	\$128,667	-\$12,898 below
2 Opportunity Youth Academy	0.437	\$161,992	+\$20,427 above
3 Abraham Lincoln High	0.414	\$139,878	-\$1,687 below
4 Luis Valdez Leadership Academy	0.413	\$89,478	-\$52,087 below
5 Yerba Buena High	0.411	\$89,478	-\$52,087 below

--- San Jose: Top 5 Least At-Risk ---

A tibble: 5 × 4

School <chr>	`Risk Score` <dbl>	`Buffer Income` <chr>	`vs. City Median` <chr>
1 Piedmont Hills High	0.176	\$173,330	+\$31,765 above
2 Leland High	0.182	\$240,028	+\$98,463 above
3 Willow Glen High	0.194	\$183,166	+\$41,601 above
4 Santa Teresa High	0.195	\$180,824	+\$39,259 above
5 Evergreen Valley High	0.200	\$181,919	+\$40,354 above

In **Stockton**, where the average median household income is **\$76,851**, **4 out of 5** of the **Top 5 most at-risk schools** are located in school-neighborhoods with lesser household income compared to the city average, ranging from **-\$20,402** to **-\$34,028**. **3 out of 5** of the **Top 5 least at risk-schools** are located in school-neighborhoods with lower household income compared to the city average, ranging from **-\$11,493** to **-\$26,396**, and **2 out of 5** are located in school-neighborhoods with higher household income compared to the city average, ranging from **+\$12,652** to **+\$32,457**.

--- Stockton: Top 5 Most At-Risk ---

A tibble: 5 × 4

School <chr>	`Risk Score` <dbl>	`Buffer Income` <chr>	`vs. City Median` <chr>
1 Stockton High	0.363	\$42,823	-\$34,028 below
2 Health Careers Academy	0.3	\$47,801	-\$29,050 below
3 Middle College High	0.262	\$54,605	-\$22,246 below
4 Lincoln High	0.243	\$83,339	+\$6,488 above
5 Edison High	0.213	\$56,449	-\$20,402 below

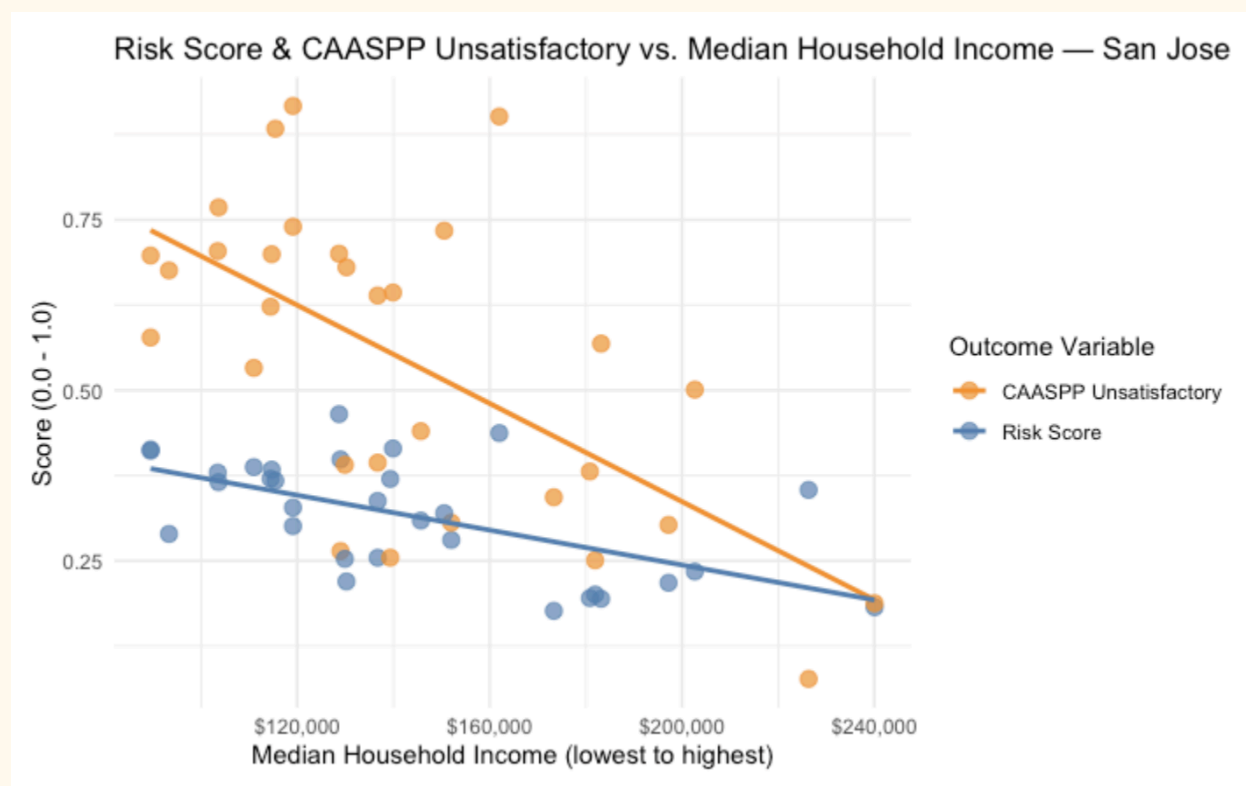
--- Stockton: Top 5 Least At-Risk ---

A tibble: 5 × 4

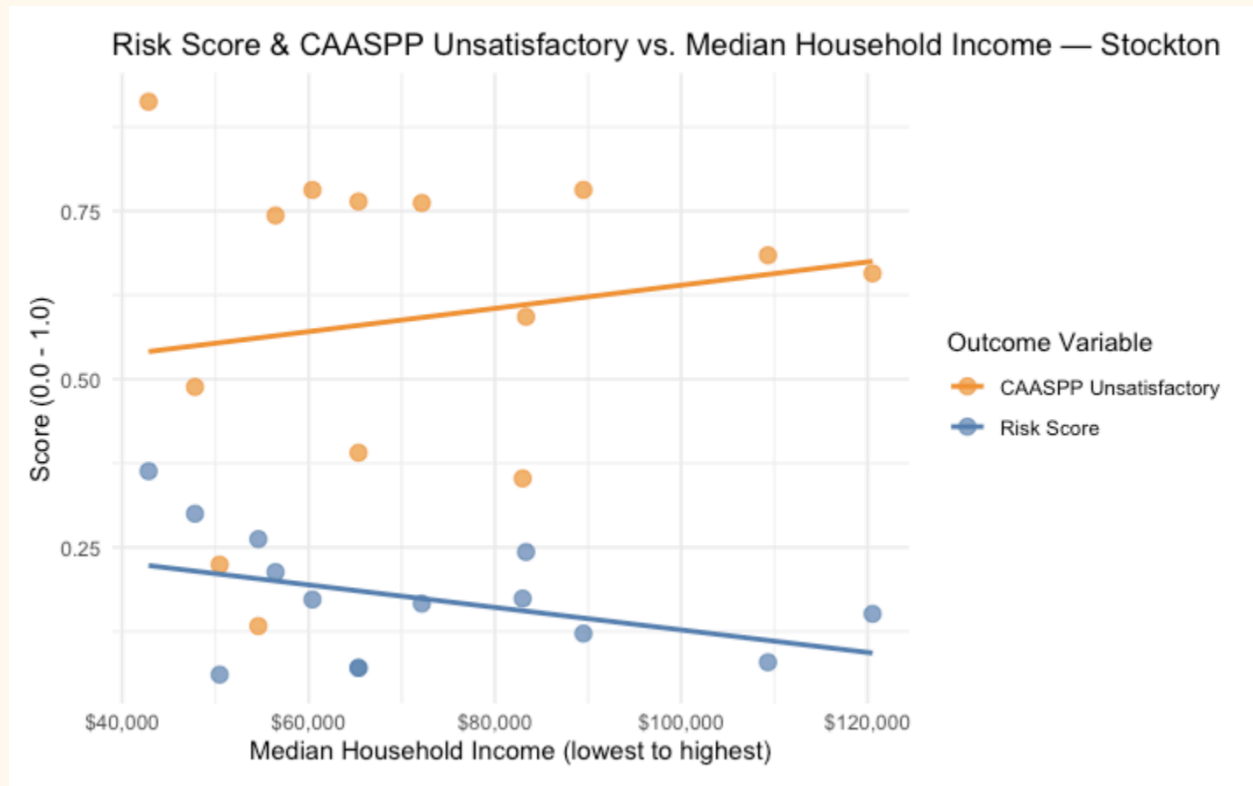
School <chr>	`Risk Score` <dbl>	`Buffer Income` <chr>	`vs. City Median` <chr>
1 Stockton Early College Academy	0.0607	\$50,455	-\$26,396 below
2 Pacific Law Academy	0.0708	\$65,358	-\$11,493 below
3 Stagg Senior High	0.0708	\$65,358	-\$11,493 below
4 Weston Ranch High	0.0791	\$109,308	+\$32,457 above
5 Cesar Chavez High	0.122	\$89,503	+\$12,652 above

Across the income range for both cities' school-neighborhoods, **lower income** is **mostly negatively related** to **greater Risk Score**, while **higher income** is **mostly negatively related** to **lesser Risk Score**. **San Jose experiences the larger wealth gap** across its Top 5 most and least at-risk school-neighborhoods, while **Stockton experiences generally lower income**, with lesser variable range between each school, for its Top 5 most and least at-risk school-neighborhoods.

The below scatter plot for San Jose high schools represents the **negative relationship** between **income, Risk Score, and % Unsatisfactory CAASPP scores** on a **0.0 to 1.0 scale**, where higher income reflects lower risk as well as a lesser percentage of students who fail to meet standard on the CAASPP. *Code for this graph is viewable on lines 1301 through 1362 in the .Rmd file in [GitHub](#).*



On the other hand, the scatter plot with data for Stockton high schools represents a **negative relationship** between **income and Risk Score**, but a **positive relationship** between **income and % Unsatisfactory CAASPP**. However, this is not statistically robust as, with a tiny sample size of just 13 Stockton high schools, one or two schools that deviate from the rest of the data set can flip the slope of the graph entirely. In fact, **Stockton Early College Academy** is a selective, higher-performing school that sits in a low-income downtown area, therefore reflecting a **lower % Unsatisfactory CAASPP** despite income trends predicting otherwise. *Code for this graph is viewable on lines 1386 through 1447 in the .Rmd file in [GitHub](#).*



CONCLUSION: TAKEAWAYS, LIMITATIONS, HOPES FOR THE FUTURE

Takeaways

Based on my research, transportation and mobility conditions and inequality does impact educational access and engagement as reflected in differences in academic performance (i.e. standardized test scores). This particularly impacts students in under-served communities (i.e. lower income areas), with poor transportation safety and mobility access most likely exacerbating risk indicators associated with the school-to-prison pipeline (i.e. poor academic performance).

While we can likely use mobility and transportation conditions together (MAI and TSI → Risk Score) to determine this risk of poor academic performance, this relationship proves stronger in larger, more populated cities (e.g. San Jose has a stronger relationship in its variables with the city's available data compared to a weaker relationship represented by Stockton data). In addition, transportation safety seems to be the stronger predictive variable of risk compared to

mobility access. Mobility access does not always positively correlate to higher income, lesser risk, or better academic performance within an area.

Limitations

- The **small number of schools** (46 total, 32 in San Jose, 14 in Stockton) was not the ideal sample size, as the research could have benefited from more accurate interpretations with larger sample sizes.
 - While there is a positive relationship between Risk Score and poor academic performance, we would need a lot more schools to prove a stronger relationship.
- The **crime exposure data** accounted for in the Transportation Safety Index reflects only crime events that have been reported by police departments, which may leave some information out even if events occurred in actual environments.
- The **crash data** accounted for in the Transportation Safety Index could only account for a ½ mile radius around the school, which does not fully capture the 1-mile school-neighborhood buffer zone.
- We **cannot see** individual student's transportation routes in our data, nor if students have stopped using public or district transportation options because of crime in an area, which does occur anecdotally.
- **Transportation and mobility infrastructure is just one lens** to causes of the school-to-prison pipeline, and a holistic understanding of pipeline risk indicators requires an **intersectional analysis** across various social, environmental, economic, policy, and further systemic causes.

Hopes for the Future

- **Including more schools and larger cities** in my data, potentially repeating this research project with a significantly larger data set, could lead to stronger insights.
- **Chronic absenteeism** data was not available for this research project, despite being my first choice and the most closely linked to mobility access and transportation safety. I would be very interested in expanding and repeating this project using chronic absenteeism as my outcome variable and risk indicator of interest.
- I am additionally curious how the relationships in this version of the research project would compare to relationships found using **suspension or graduate rate data**.

Lastly, this project helps answer which school communities may need infrastructure investment before youth are pushed into punishment-based systems for poor academic performance and other outcomes that are actually risk indicators along the school-to-prison pipeline. As a potentially strong policy framework, future iterations of this research can likely identify neighborhoods where “performance / school engagement problems” may actually reflect structural barriers, identify schools needing transportation support, prioritize attendance interventions, allocate community resources, expand school transit access, and target violence prevention infrastructure.